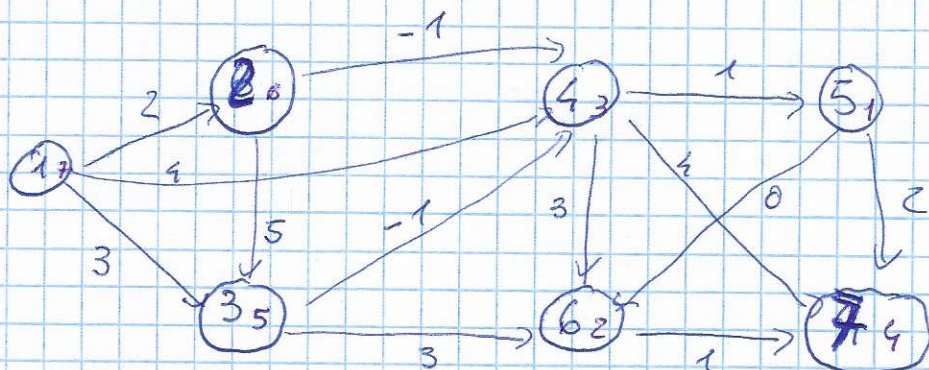
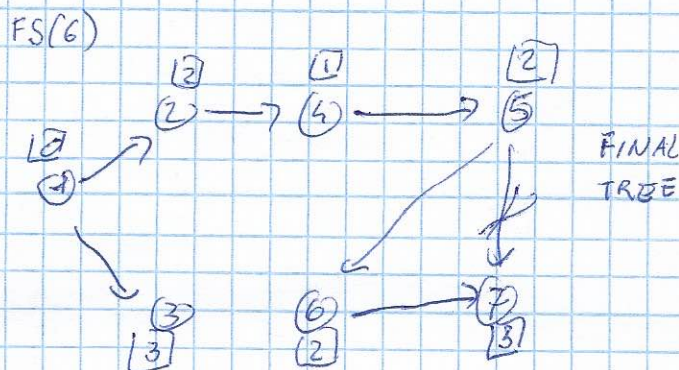
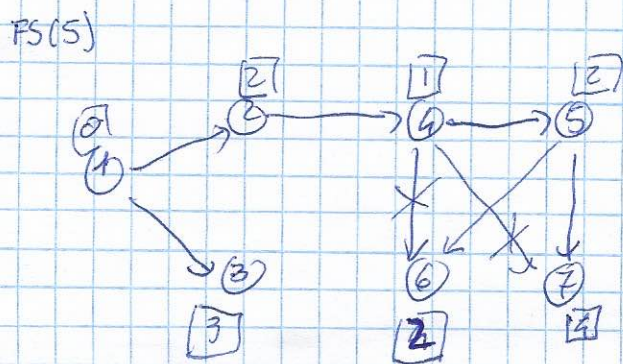
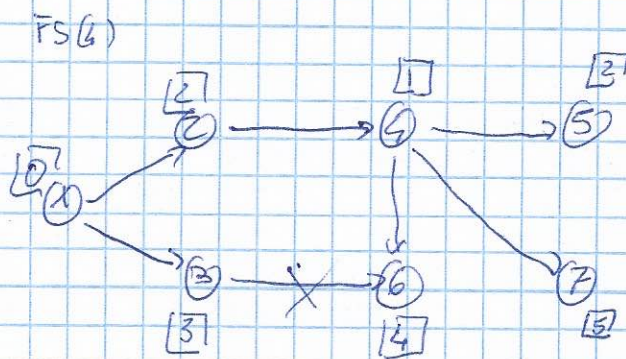
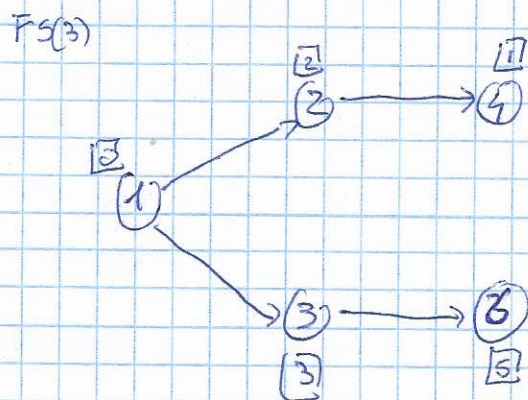
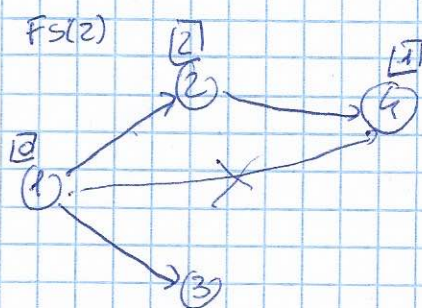
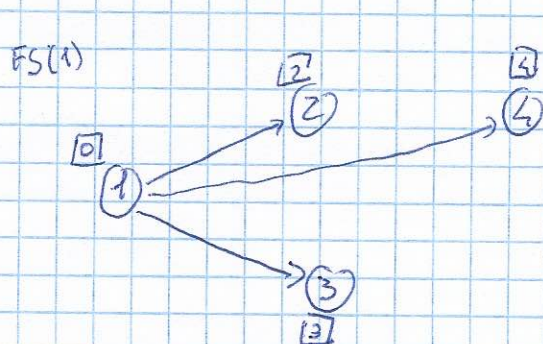


The graph nodes can be renumbered according to the Topological order, thus G is acyclic.

The renumbered graph is the following one



Regardless of the negative costs, we can apply SPT_Acyclic whose complexity, is linear in the number of arcs



$$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ violates constraint 1: } 0 \leq -3 \quad (2)$$

The auxiliary problem is

$$\max -u_1$$

$$\text{AP: } -3x_1 - 2x_2 - u_1 \leq -3$$

$$-10x_1 + 6x_2 \leq 9$$

$$2x_2 \leq 7$$

$$2x_1 + 2x_2 \leq 11$$

$$-u_1 \leq 0$$

A trivial initial solution for AP is $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $u_1 = -b_1$,
 thus $\bar{x} = \begin{bmatrix} 0 \\ 0 \\ +3 \end{bmatrix}$ $I(\bar{x}) = \{1\}$ $A_1 = [-3 -2 -1]$

The conditions for the existence of a growing feasible direction in $\bar{x}$ are

$$\begin{cases} c^T \xi > 0 \\ A_i^T \xi \leq 0 \end{cases} \text{ where } \xi = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} \text{ thus } \begin{cases} -\xi_3 > 0 \\ -3\xi_1 - 2\xi_2 - \xi_3 \leq 0 \end{cases}$$

$$\xi = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \text{ satisfies the condition } \begin{matrix} > 0 \\ 0 \leq 0 \end{matrix}$$

Let us compute λ

$$A_2 \xi = [10 \ 6 \ 0] \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = 6 \quad \frac{(b_2 - A_2 \bar{x})}{A_2 \xi} = \frac{3}{2} \rightarrow \lambda$$

$$A_3 \xi = [0 \ 2 \ 0] \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = 2 \quad \frac{(b_3 - A_3 \bar{x})}{A_3 \xi} = \frac{7}{2}$$

$$A_4 \xi = [2 \ 2 \ 0] \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = 2 \quad \frac{(b_4 - A_4 \bar{x})}{A_4 \xi} = \frac{11}{2}$$

$$A_5 \xi = [0 \ 0 \ -1] \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = 2 \quad \frac{(b_5 - A_5 \bar{x})}{A_5 \xi} = \frac{3}{2} \rightarrow \lambda$$

$$x = \begin{bmatrix} 0 \\ 0 \\ +3 \end{bmatrix} + \frac{3}{2} \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3/2 \\ 0 \end{bmatrix}$$

We note that item 6 has value 1 and weight - 5 (3)

thus we can fix variable x_6 to 1 the problem becomes:

$$1 + \max 12x_1 + 12x_2 + 13x_3 + 6x_4 + 2x_5$$

$$P_0: 4x_1 + 5x_2 + 6x_3 + 3x_4 + 2x_5 \leq 10$$

$$x_i \in \{0, 1\} \quad i=1..5$$

We sort the items by ratio value/weight

$$\frac{12}{4} > \frac{12}{5} > \frac{13}{6} > \frac{6}{3} > \frac{2}{2}$$

We apply the greedy to find a feasible solution

$$x_1 = 1 \quad 10 \rightarrow 6 \quad x_2 = 1 \quad 6 \rightarrow 1 \quad x_3 = x_4 = x_5 = 0 \quad \text{value } 24 \\ \text{infeasible sol.}$$

ub of P_0

$$x_1 = 1 \quad x_2 = 1 \quad x_3 = 1/6 \quad \text{ub} = 12 + 12 + \frac{13}{6} = 26 + \frac{1}{6} \rightarrow 26$$

Branch on x_3

$$P_1: P_0, x_3 = 1$$

$$13 + \max 12x_1 + 12x_2 + 6x_4 + 2x_5$$

$$4x_1 + 5x_2 + 3x_4 + 2x_5 \leq 4$$

$$x_i \in \{0, 1\} \quad i=1, 2, 4, 5$$

since the weight of item 2 > 4 we can fix x_2 to 0

ub of P_1

$$x_1 = 1 \quad \text{value } 13 + 12 = 25 \quad \text{new infeasible sol.} \\ \text{feasible}$$

$$P_2: P_0, x_3 = 0$$

$$\max 12x_1 + 12x_2 + 6x_4 + 2x_5$$

$$4x_1 + 5x_2 + 3x_4 + 2x_5 \leq 10$$

$$x_i \in \{0, 1\} \quad i=1, 2, 4, 5$$

ub of P_2

$$x_1 = 1 \quad x_2 = 1 \quad x_4 = \frac{1}{3} \quad \text{value } 12 + 12 + \frac{6}{3} = 26 > \text{infeasible}$$

Branch on x_4

$$P_3: P_2, x_4 = 1$$

$$6 + \max 12x_1 + 12x_2 + 2x_5$$

$$4x_1 + 5x_2 + 2x_5 \leq 7$$

$$x_i \in \{0, 1\} \quad i=1, 2, 5$$

$$ub \quad x_1 = 1 \quad x_2 = \frac{2}{5}$$

$$value \quad 6 + 12 \cdot \frac{2}{5} = 22 + \frac{4}{5} \sim 22$$

$\leq$ incumbent
stop

$$P_4: P_2, x_4 = 0$$

$$\max 12x_1 + 12x_2 + 2x_5$$

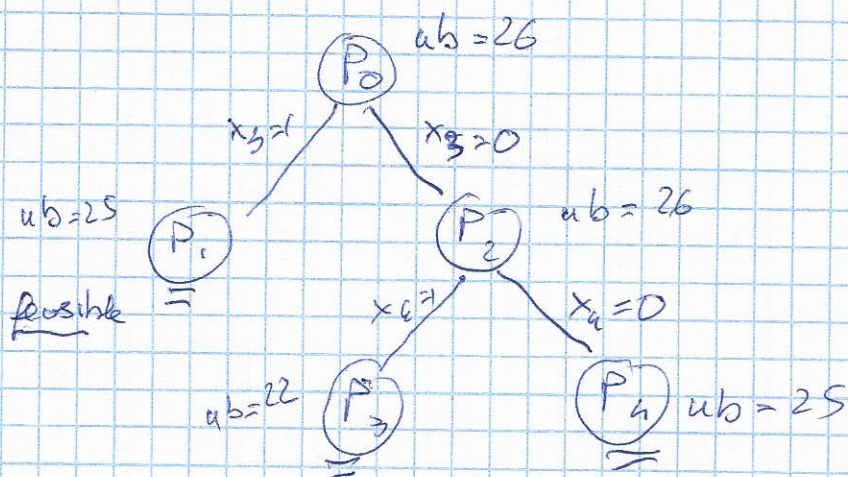
$$4x_1 + 5x_2 + 2x_5 \leq 10$$

$$x_i \in \{0, 1\} \quad i=1, 2, 5$$

$$ub \quad x_1 = 1 \quad x_2 = 1 \quad x_5 = 1/2$$

$$value = 24 + 1 = 25$$

$\leq$ incumbent
stop



the optimal solution is

$$x_1 = 1 \quad x_3 = 1 \quad x_2 = x_4 = x_5 = 0 \quad value = 25$$

(4)

let us suppose $Cx' = Cx'' = z$

A convex combination of x' and x'' is

$$\bar{x} = \lambda x' + (1-\lambda)x'' \quad \text{with } \lambda \in [0, 1]$$

$$\begin{aligned} C\bar{x} &= \lambda Cx' + (1-\lambda)Cx'' = \\ &\lambda z + (1-\lambda)z = z \end{aligned}$$

thus also $\bar{x}$ has value z

since $\bar{x}$ is a convex combination of two feasible points, it is also feasible

$\Rightarrow \bar{x}$ is optimal

(5)

Sets $I = \{1 \dots n\}$
 $O = \{1 \dots n\}$

Parameters

a_i time $i \in I$

s_i station $i \in I$

b_j time $j \in O$

s_j station $j \in O$

c_{ij} travel time $i \in I \quad j \in O$

Variables

$x_{ij} = \begin{cases} 1 & \text{input stamp } i \text{ matched with output stamp } j \\ 0 & \text{otherwise} \end{cases}$

$\forall i \in I \quad \forall j \in O$

Additional parameters

Let $d_{ij} = |c_{ij} - (b_j - a_i)|$ that can be computed in advance

The problem is

$\min \max d_{ij} x_{ij}$

$$\sum_{j \in O} x_{ij} = 1 \quad \forall i \in I$$

$$\sum_{i \in I} x_{ij} = 1 \quad \forall j \in O$$

To linearize the objective function let us introduce the additional dummy variable D

and the constraints

$$D \geq d_{ij} x_{ij} \quad \forall i \in I \quad \forall j \in O$$

$$E = \{ (i, j, h) \mid \begin{array}{l} i = 1 \dots n-2 \\ j = i+1 \dots n-1 \\ h = 1 \dots n \end{array} \}$$

(6)

E is the set of all different Terms

$F = \{ \text{subset of terms without common elements} \}$

$$\text{Best}(S) = \text{argmax} \{ (c_{ij} + c_{jh} + c_{ih}) : (i, j, h) \in S \}$$

$\text{ind}(X, (i, j, h)) = \text{true}$ if i, j and j and h do not appear in any Term in X